Do database magnetic labels hold up? Screening Dzyaloshinskii–Moriya magnets

1 Harmonize & filter

64,635
Records, 7 databases

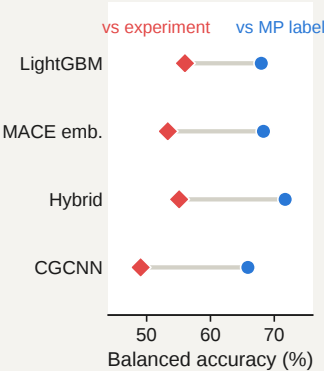
47,614
Net moment (FM/FiM)

19,280
Noncentrosymmetric

1,864
Heavy spin–orbit element

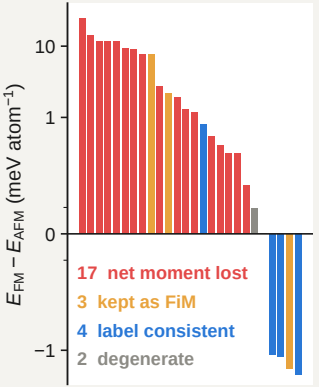
1,195
Hull ≤ 100 meV atom^{-1}

2 Classify ordering

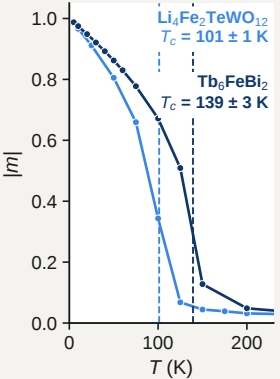


LightGBM screen: 67.4% acc., 68.0% bal.
GNNs fit MP labels better, not experiment

3 DFT on 26 MP labels



4 Spin models



DFT J and $\mathbf{D}$ ($|D|/J \approx 5.6\%$)
→ atomistic spin dynamics + FSS

17 of 26 net-moment labels lost in DFT → ordering labels are a screening prior, not ground truth